IDRIS RAYAN CHAKIB

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Education

University of M'Hamed Bougara Boumerdes — **Boumerdes, Algeria**

Master of Science in Software Engineering and Data Processing — **Sep 2024 - June 2026**

- **Thesis Title:** "AI Pipeline for Genetic Variant Analysis and Virtual Therapeutic Screening" (In partnership with Institut Pasteur d'Algérie).
- **Academic Distinctions:** Thésis Grade : 18/20 (Graduated with Highest Honors / Congratulations of the Jury).
- **Startup Path (Decree 1275):** Developed the project's core framework as an innovative startup. Startup Grade: 19/20 | Awarded the Official National Startup Certificate.
- **Key Coursework:** Network and communication protocols, Cybersecurity, Distributed Systems, Software Architecture, Data Mining.

University of M'Hamed Bougara Boumerdes — **Boumerdes, Algeria**

Bachelor of Science in Software Engineering and Information Systems — **Sep 2021 - June 2024**

- **Academic Standing:** Class Rank: 39 / 182 (Top 21.4% of the cohort / Equivalent to ECTS Grade B).
- **Capstone Project Title:** "Fraud Monitoring Web Solution with Machine Learning" (In partnership with Ooredoo Algeria).
- **Project Distinction:** Grade 16/20.
- **Key Coursework:** Object-Oriented Programming, Calculus, Linear Algebra, Probability & Statistics, Databases**.**

Research Publication

- Benhenni O., Gaceb D., Touazi F., Rouibi B., Ismail A., Kebour W., **Idris R. C.**, Galleze A., Benidir M. "**SmartADN: An End-to-End Deep Learning Pipeline for Variant Pathogenicity Classification and Mutation-Aware Therapeutic Screening in the North African Context.**" Submitted to the IEEE International Symposium on iNnovative Informatics of Biskra (**IEEE ISNIB'26**), 2026. (Paper ID: 101 Currently Under Peer-Review / Forthcoming).
- Benhenni O., Gaceb D., Touazi F., Kebour W., **Idris R. C**., Rouibi B., Ismail A. "**An End-to-End Approach for SNV Pathogenicity Prediction via Progressive Fine-Tuning of Nucleotide Transformer-v2.**" Published and presented at the 3rd National Conference on Applications of Artificial Intelligence (**A2I'26**), 2026. (Official Certificate of Participation Awarded as **Presenting Author**).

Professional Experience

Institut Pasteur d'Algérie — **Algiers, Algeria**

Data Processing & AI Researcher Intern / Computational Biology — **Feb 2026 – June 2026**

- Co-designed SmartADN, an integrated pipeline coupling a sequence-only pathogenicity classifier with a mutation-aware therapeutic screening module.
- Fine-tuned the **Nucleotide Transformer-v2** genomic foundation model (500M parameters) using a layer-wise unfreezing strategy under a strict gene-split out-of-distribution (OOD) protocol.
- Achieved a near-perfect AUC-ROC of **0.987** and **94%** overall accuracy on entirely held-out clinically critical genes (BRCA1, BRCA2, TP53) without relying on population allele-frequency annotations.
- Engineered the protein backbone integration for **AEGIS-GT**, mapping conformational perturbations via the pointwise difference of ESM-2 embeddings ($\Delta e = eMUT - eWT$).
- Implemented a **bi-phasic training strategy** using a linear zero-initialised adapter to successfully bridge biophysical Silver-Standard descriptors with experimental Gold-Standard affinity data.

Ooredoo Algeria — **Algiers, Algeria**

Software Engineering Intern — **Feb 2024 – June 2024**

- Ingested and preprocessed large-scale Call Detail Records (CDRs) datasets to extract telecom fraud patterns, optimizing data workflows for real-time anomaly detection.
- Designed the end-to-end distributed system architecture using UML diagrams (including deployment diagrams) to ensure scalable infrastructure deployment.
- Developed a secure, full-stack web platform with Role-Based Access Control (RBAC) to streamline analytical workflows for administrators and analysts.
- Built and deployed a feedforward neural network model for fraud detection, validating infrastructure performance with complex real and simulated data streams.

Projects

Multi-Robot Autonomous 2D Swarm Exploration

- Engineered a multi-robot swarm mapping system from scratch using a continuous Teacher-Student reinforcement learning strategy paired with the A* pathfinding algorithm.
- Coded the entire Deep Q-Network (DQN) architecture, backpropagation loops, and neural layers using raw Python/NumPy.

Nexus: Decentralized P2P Encrypted Chat System

- Designed a robust peer-to-peer distributed framework featuring dynamic private chat rooms, logical isolation, and multi-threaded file transfer protocols.
- Implemented physical concurrency controls and data consensus layers using Lamport Logical Clocks, custom Byzantine-resilient voting systems, and multi-layered Fernet cryptographic encryption via Tkinter GUI.
- Dealt with complex infrastructure stacks involving distributed systems, concurrency control, and socket-based network protocols.

Moogle: Natural Language Processing Search Engine

- Built a highly optimized localized text search engine integrating foundational Natural Language Processing (NLP) preprocessing pipelines.
- Developed custom algorithmic structures for tokenization, automated stop-word elimination, Porter-style stemming/unstemming, and contextual Bi-gram language modeling.

Leadership Experience

SmartADN Startup Project — **Boumerdes, Algeria**
Co-Founder & Lead AI Engineer — **Sept 2025 – Present**

- Selected for the prestigious Algerian National Student-Entrepreneur program under Ministerial Decree 1275, demonstrating academic and technical leadership.
- Architected and developed AEGIS-GT, a custom deep learning model designed for advanced genomic analysis and core startup technology.
- Successfully pitched the technical design and AI model architecture to academic committees, securing the official National Startup Certificate.

NASA International Space Apps Challenge — **Global / Algiers Cohort**
Team Leader & Core Developer

- Led a multidisciplinary team of student engineers and peers during the 48-hour global hackathon to address complex astronomical challenges.
- Co-designed and engineered a 3D interactive web-based Orrery application to simulate celestial body trajectories using open-source NASA telemetry data.
- Managed project workflows, delegated tasks among university peers under strict time constraints, and delivered the functional backend framework, awarded the **Galactic Problem Solver Certificate**

Additional Credentials & Activities

Professional Certifications — **2025 – 2026**

- **Machine Learning Specialization (DeepLearning.AI):** Certified core expertise in neural networks and AI system design.
- **Google Data Analytics Professional Certificate:** Certified proficiency in data cleaning, SQL analysis, and problem-solving.

Active Competitor, Codeforces Platform — **2026 – Present**

- Current Rating: 1166 (Actively progressing toward top-tier algorithmic milestones via timed challenges.)
- **Ranked 273rd out of 3,830 active global participants (Top 7%)** in the ICPC Online Challenge 1 (powered by Huawei); engineered a custom scheduling heuristic to optimize concurrent request routing for Edge-Cloud Collaborative LLM Inference loops.

Technical Skills

- **Languages:** Java, Python, C/C++, SQL, JavaScript, HTML/CSS.
- **Cloud Platforms:** Google Cloud Platform, AWS, Microsoft Azure.
- **Frameworks:** Next, Express, FastAPI, Flask, Pytorch, Keras, Scikit-learn, Tensorflow, Hugging Face (Transformers).
- **Developer Tools:** Linux/UNIX, Git, Docker, Kubernetes, Node.js, VS Code, Visual Studio, Pycharm, IntelliJ, Eclipse.
- **Libraries:** REST APIs, Microservices, React, pandas, Numpy, Matplotlib, seaborn.